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A SHAP-based framework for visualizing regional heterogeneity in pavement deterioration

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Abstract

This study focuses on regional heterogeneity in road-pavement deterioration. Accurate prediction of deterioration is essential for efficient maintenance planning. However, traditional statistical deterioration prediction approaches often have difficulty in capturing the complex nonlinear interactions among deterioration factors and the regional heterogeneity of deterioration, particularly when how regional conditions around road pavement affect the deterioration is uncertain. This study developed a SHAP-based framework for visualizing regional heterogeneity in pavement deterioration. Regional dummy variables are included in the model, and their SHAP values are interpreted as distributions to quantify the influence of regional characteristics on pavement deterioration. A case study using approximately 430,000 inspection records from Japan's national highways validates the framework. The results reveal that regional location is less influential than primary factors such as time in service and traffic volume, yet its impact is comparable to structural characteristics such as pavement material. Western Japan exhibited overall low deterioration speed, while eastern Japan, especially the Sea of Japan coastal regions, showed high deterioration speed. Overall, the approach visualizes regional heterogeneity in detail and may support region-adaptive maintenance planning.

Keywords Pavement deterioration, Asset management, Regional heterogeneity, SHAP, Neural network

1 Introduction

Infrastructure such as road pavements plays a vital role in maintaining the safety and functionality of society. In developed countries, where much of the infrastructure has significantly aged since its initial construction, managing and renewing them within the limits of constrained public budgets has become an increasingly urgent and complex challenge. Meeting this challenge requires allocating maintenance resources at appropriate times and locations. To support such decision-making, it is crucial to develop deterioration prediction model that can accurately predict future deterioration of the road pavement based on available data and plausible assumptions regarding the time in service, traffic volume, and regional conditions of infrastructure [1].

Accurately predicting pavement deterioration remains challenging. A major reason is regional heterogeneity: both the base deterioration speed and the effects (and interactions) of factors (such as traffic demand and climate) differ across regions. Together with uneven inspection records and limited information on infrastructure usage—situations commonly encountered in practice—this heterogeneity complicates estimation and limits out-of-region transferability. In response, recent studies have developed prediction models that explicitly capture regional heterogeneity [2, 3].

However, analyses that focus on regional heterogeneity often rely on representative statistics such as regional

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averages, potentially overlooking the diversity that exists within each region. In particular, when a region is assumed to span a substantial geographical area, simply being "located within a region" does not necessarily exert a uniform influence on the deterioration. For example, certain usage conditions may accelerate deterioration, while others may have a mitigating effect. Such diversity in deterioration is likely to arise from the interaction between regional location and individual deterioration factors, as well as from the nonlinear nature of these influences.

Conventional modeling approaches, including standard regression models and mixed Markov hazard models, have contributed significantly to the understanding of infrastructure deterioration. These models typically assume a linear relationship between deterioration factors and deterioration speed, which offers interpretability and analytical tractability. However, such assumptions may limit their ability to fully capture the complex, nonlinear interactions observed in diverse real-world contexts. In contrast, machine learning models—which have advanced rapidly in recent years—provide a flexible framework capable of representing both interaction effects and nonlinear relationships. Furthermore, the application of SHAP (SHapley Additive exPlanations), a widely used method in the field of explainable artificial intelligence, enables the influence of individual deterioration factors and regional location on deterioration speed to be evaluated quantitatively and intuitively in the form of distributions.

This study focuses on pavement inspection data collected from national highways across Japan and develops a deterioration prediction model using a neural network approach. By applying SHAP to the trained model, this study proposes a framework for quantitatively capturing regional heterogeneity in deterioration behavior as region-specific distributions.

Section 1 has presented the background and objectives of this study. Section 2 provides a review of the relevant literature. Section 3 describes the proposed framework in detail. Section 4 conducts a case study to demonstrate the applicability and effectiveness of the proposed framework. Finally, Section 5 concludes the study.

2 Literature review

Three major statistical approaches have been developed for modeling the deterioration processes of infrastructure: regression-based models [4], Markov hazard models [5], and machine learning-based models [6]. These studies commonly aim to quantify the progression of deterioration and improve predictive accuracy, while also seeking to identify deterioration factors and elucidate underlying mechanisms through modeling.

Many of these statistical approaches require large volumes of high-quality deterioration data for adequate to achieve adequate model performance. However, in practice, such data is not always readily available. For instance, deterioration-related data are often insufficiently accumulated in developing countries [2]. Even in developed countries, some infrastructure systems—such as sewer networks—lack comprehensive inspection records because large-scale inspections are difficult to conduct [3]. Lack of the high-quality deterioration data for achieving accurate deterioration prediction model has been reported worldwide [2, 3]. To address these limitations, a common strategy is to integrate multiple sources of data to compensate for the lack of complete individual records. Llopis et al. [7] demonstrated that pavement deterioration is influenced by the combined effects of pavement structure, traffic volumes, and weather conditions, highlighting the importance of accounting for interactions among deterioration factors. Nevertheless, it remains challenging to accurately capture the characteristics and interactions of deterioration factors across regions, given that climatic conditions, traffic volumes, and construction practices often vary widely.

Against this background, increasing attention has been given to analytical methods capable of incorporating complex factor interactions and regional variation. One such method is SHAP, a technique from the field of explainable AI that enables the visualization and quantification of the contribution of each variable to a machine learning model's prediction. Recent studies have applied SHAP to explore causal factors behind road cave-ins in urban areas [8]. However, no existing research has analyzed regional heterogeneity in the progression of deterioration, treating it as a gradual process rather than a binary outcome. This study applies SHAP to a deterioration prediction model to visualize and analyze the characteristics and diversity of regional heterogeneity.

3 Framework

3.1 Overview of the proposed framework

This study proposes a framework for quantitatively and comprehensively characterizing regional heterogeneity in infrastructure deterioration by representing it as a distribution across regions. The framework involves applying SHAP, a technique developed in explainable AI (XAI) to a deterioration prediction model capable of capturing nonlinear relationships and interactions among explanatory variables. By doing so, the contribution of each explanatory variable to the predicted deterioration level can be visualized as a distribution. In the proposed framework, the complex interactions and nonlinear effects between the variable "regional location" and other explanatory variables are interpreted as the diversity in how regional location affects deterioration. This diversity is extracted and represented in distributional form.

3.2 Requirements for the deterioration prediction model

The deterioration prediction model used in the proposed framework must meet two key requirements: (1) it must include several explanatory variables, including dummy variables representing regional location; and (2) it must be capable of modeling nonlinear relationships and interactions among explanatory variables in their influence on deterioration. The choice of model type is not restricted. Thus, the framework can be applied to various models, including neural networks, multiple regression models with interaction terms, and mixed Markov deterioration models that account for interactions.

3.3 Overview of SHAP

In the proposed framework, SHAP is used to analyze the influence of explanatory variables—particularly regional dummy variables—on deterioration. SHAP, proposed by Lundberg and Lee [9], is one of the most widely used techniques in the field of explainable artificial intelligence (XAI). While machine learning models have long been recognized for their high predictive performance, they have also been criticized for their “black-box” nature, which obscures the reasoning behind their outputs. Since 2017, however, rapid progress has been made in the development of techniques for interpreting machine learning models. SHAP offers a unified framework that incorporates several earlier interpretation methods, such as LIME, by applying the concept of Shapley values from cooperative game theory. Grounding SHAP in this theoretical foundation ensures desirable properties such as local accuracy, missingness, and consistency, giving the method a high degree of theoretical soundness. A brief overview of SHAP is provided below.

Let the deterioration prediction model be denoted by $f(\boldsymbol{x}_i)$. Here, $\boldsymbol{x}_i$ is the vector of explanatory variables representing target data point i, which serves as the input to the model. The vector $\boldsymbol{x}_i$ is defined as follows:

$$\boldsymbol{x}_i := \left[x_{i,1}, x_{i,2}, \cdots, x_{i,j}, \cdots, x_{i,|\boldsymbol{N}|}\right]^\top \forall i \in \boldsymbol{I} \tag{1}$$

where $\boldsymbol{N}$ denotes the set of explanatory variables, $|\boldsymbol{N}|$ is the number of variables, and $i \in \boldsymbol{I}$ indicates that this definition applies to all target data points in the dataset $\boldsymbol{I}$.

For any subset $\boldsymbol{S} \subseteq \boldsymbol{N}$, the masking operator $h_{\boldsymbol{S}}(\boldsymbol{x}_i; \boldsymbol{z})$ is defined to keep the features in $\boldsymbol{S}$ at their observed values and to substitute the remaining features with the corresponding entries of a background sample $\boldsymbol{z}$. A background sample $\boldsymbol{z} = (z_1, z_2, \cdots, z_j, \cdots, z_{|\boldsymbol{N}|})^\top$ is a reference data point used to replace the masked features during SHAP calculations. It provides a baseline for comparison, allowing the effect of including or excluding specific features to be quantified in terms of changes in the model's output. The background samples are drawn from a background distribution $p\left(\boldsymbol{Z}|\boldsymbol{x}_{i,\boldsymbol{S}}\right)$, which represents the statistical relationship between the unmasked features $\boldsymbol{x}_{i,\boldsymbol{S}}$ and the masked features. In practice, this background distribution is usually approximated using an empirical dataset such as the training data or a representative subset of it. The masking operator is formally defined as:

$$[h_{\boldsymbol{S}}(\boldsymbol{x}_i; \boldsymbol{z})]_j := \begin{cases} x_{i,j} j \in S \\ z_j j \notin S \end{cases} \forall j \in \boldsymbol{N}, i \in \boldsymbol{I} \tag{2}$$

where $x_{i,j}$ is the j-th feature of the target data point and z_j is the corresponding feature value of the background sample.

The model output for a masked input is defined as the expected prediction over the background distribution. This represents the expected model output when the masked features are replaced by values sampled from the background distribution:

$$f(h_{\boldsymbol{S}}(\boldsymbol{x}_i)) := \mathbb{E}_{\boldsymbol{Z}\sim p(\boldsymbol{Z}|\boldsymbol{x}_{i,\boldsymbol{S}})}[f(h_{\boldsymbol{S}}(\boldsymbol{x}_i; \boldsymbol{Z}))] \tag{3}$$

where $\boldsymbol{x}_{i,\boldsymbol{S}}$ denotes the sub-vector of $\boldsymbol{x}_i$ indexed by $\boldsymbol{S}$, and $\boldsymbol{Z}$ is a random background vector drawn from the background distribution.

In theory, Eq. (3) uses the full conditional distribution $p\left(\boldsymbol{Z}|\boldsymbol{x}_{i,\boldsymbol{S}}\right)$ which accurately models the statistical dependencies between masked and unmasked features. However, modeling such dependencies can be computationally expensive or infeasible in practice. Therefore, a common practical simplification is to assume feature independence, so that the conditional distribution $p\left(\boldsymbol{Z}|\boldsymbol{x}_{i,\boldsymbol{S}}\right)$ simplifies to the marginal distribution $p(\boldsymbol{Z})$. Furthermore, in practice, this expectation in Eq. (3) is approximated using a finite set of background samples indexed by the set $\boldsymbol{M}$. Let $\boldsymbol{M}$ be the index set of background samples, and let $\boldsymbol{z}^{(m)}$ denote the m-th realization sample drawn from the background distribution. The empirical approximation is then given by:

$$f(h_{\boldsymbol{S}}(\boldsymbol{x}_i)) \approx \frac{1}{|\boldsymbol{M}|} \sum_{m \in \boldsymbol{M}} f\left(h_{\boldsymbol{S}}\left(\boldsymbol{x}_i; z^{(m)}\right)\right) \tag{4}$$

where $|\boldsymbol{M}|$ is the cardinality of the index set $\boldsymbol{M}$. In practice, the background samples $\left\{\boldsymbol{z}^{(m)}\right\}_{m \in \boldsymbol{M}}$ are often taken directly from the training dataset or from a randomized subset of it.

Finally, the SHAP value $\phi_{i,j}$ of explanatory variable j for data point $\boldsymbol{x}_i$ is defined as:

$$\phi_{i,j} := \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (f(h_{S \cup \{j\}}(x_i)) - f(h_S(x_i))) \quad (5)$$

where the summation is taken over all subsets of the explanatory variable set that do not contain j. This equation computes a weighted average of the marginal contribution of variable j to the model prediction, evaluated across all possible subsets of variables.

Note that assuming feature independence in SHAP estimation is a practical and commonly adopted choice for computational efficiency and scalability, and we follow this practice in the present study. However, when explanatory variables are correlated, the attributions to individual variables may be over- or under-estimated. For example, in the context of pavement deterioration analyzed here, regions with higher traffic volumes may tend to have earlier construction timing and consequently longer years-in-service. Under such structural dependencies, the estimated contributions of traffic volume and years-in-service to deterioration can become unstable, and results should therefore be interpreted with caution.

3.4 Estimation method for SHAP

Calculating SHAP values exactly as defined in Eq. (5) results in computational complexity that increases exponentially with the number of explanatory variables. Therefore, in practical applications, it is common to reduce the number of background data points and target data points through sampling-based approximation methods. In the context of the proposed framework, it is essential to ensure that a sufficient number of target data points are included for each region in order to accurately capture the distributional characteristics of region-specific SHAP values. To achieve this, the background data, which represent the standard state of deterioration, are obtained through random sampling from the entire training dataset. For the target data points, the required number of samples is randomly drawn from the training dataset for each region so that the shape of the SHAP value distribution can be appropriately characterized.

In addition, several approximation techniques that leverage the model structure have been proposed. For example, in the case of neural network (NN) models, a method called DeepSHAP, which applies the principles of backpropagation, is commonly used to efficiently approximate SHAP values.

3.5 Analytical method for SHAP

To analyze the distribution of SHAP values across regions, two methods are proposed.

The first method, global sampling, aims to capture global deterioration trends by computing SHAP values based on target data points randomly sampled from the entire training dataset. When summary plots are applied to these results, the magnitude and direction of each explanatory variable's influence on deterioration can be visualized, enabling a comprehensive understanding of the principal factors contributing to pavement deterioration at the national scale.

The second method, region-specific sampling, is designed to evaluate region-specific deterioration characteristics. In this framework, SHAP values are computed using only the data points belonging to each region as target data, and the resulting SHAP distributions are analyzed in detail. Specifically, the SHAP values associated with the dummy variable corresponding to each target region are extracted, and their distributions are visualized using histograms. In addition, statistical indicators such as the mean, quartiles, skewness, and variance of the SHAP value distributions are calculated and visualized to quantitatively assess and compare the propensity for deterioration across regions. Importantly, a common background dataset is used for SHAP value computation across all regions, ensuring that comparisons are made on a consistent and uniform scale.

4 Case study

This section first describes the data and implementation methods used in the case study of this research. It then analyzes the results of the case study to evaluate the effectiveness of the proposed framework.

4.1 Data used

This study conducted a case study focusing on Japan's national highways under Japanese government's direct control. The data items used are summarized in Table 1. Multiple public databases were utilized to obtain and compute these data items.

First, data on pavement specifications and deterioration conditions were obtained from the *National Road Facility Inspection Database* (xROAD) [10], a platform that collects and gradually releases inspection data on road infrastructure across Japan. This study uses the

Table 1 List of data items

Data Item	Remarks
Pavement Condition	
Months in Service	
24h Traffic Volume (Heavy Vehicles)	
24h Traffic Volume (Light Vehicles)	
Travel Speed (Congested)	
Travel Speed (Uncongested)	
Surface Layer Material	One-hot encoded for 9 types
Surface Layer Thickness	
Regional Mesh Code	One-hot encoded for 81 regions

"Pavement Inspection Data by Record Unit Section", which contains pavement specifications and inspection results recorded for each 10-m segment by lane on national highways. The dataset was retrieved between January 8 and 10, 2025.

Traffic volume and travel speed data were taken from the *General Traffic Volume Survey*, part of the 2021 *National Road and Street Traffic Survey* [11]. Specifically, data were extracted from the "Site-based Basic Tables" corresponding to national highways, and were integrated with the aforementioned inspection dataset. The unit of integration was aligned with the pavement data—namely, 10-m segments by lane. Only travel lanes were included; overtaking lanes and turning lanes were excluded.

The preprocessing applied to each data item is summarized below. Traffic volume was evenly distributed across lanes based on the number of lanes. For single-lane roads, the combined traffic volume of both directions was used. Travel speed was assigned by direction, regardless of the number of lanes; for single-lane roads, the average of speeds in both directions was used.

Pavement condition was rated using a three-level scale (I–III) in accordance with the *Pavement Inspection Guidelines* [12] published by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT). Level I indicates minimal deterioration and a sound surface, Level II indicates moderate deterioration, and Level III indicates that the surface either exceeds the management threshold or is expected to do so in the near future. The time in service was calculated as the difference between the inspection date and the construction date of the first pavement layer. If the inspection date preceded the construction date, the record was deemed erroneous and excluded. Region codes were assigned based on the primary mesh system, which divides Japan into roughly 80 km grids, and were determined using the latitude and longitude of the start point of each record unit section. Pavement material for the first layer was reclassified into 9 categories (e.g., coarse-graded asphalt mixture, fine-graded asphalt mixture, dense-graded asphalt mixture, porous/open-graded asphalt mixture, other asphalt mixtures, cement concrete, concrete block, stabilized materials, others), and each data point was labeled accordingly. Pavement thickness values of 10 cm or more were treated as outliers and excluded. Region codes and material types were converted into dummy variables using one-hot encoding. Other numerical variables were standardized using IQR-based scaling, which involves dividing the deviation from the median by the interquartile range (difference between the first and third quartiles), to normalize the scale and reduce the impact of outliers.

Regarding the surface layer thickness, the inspection data exhibited a substantial number of entries in the range of 10 to 250 cm, indicating possible unit input errors during data entry (i.e., confusion between centimeters and millimeters). Therefore, all records in which the surface layer thickness was 10 cm or greater were treated as outliers and excluded. Additionally, all data points with missing items in any field were excluded. Furthermore, to ensure an accurate assessment of regional deterioration patterns, this study also excluded grid cells (meshes) with fewer than 6,000 data points. After these preprocessing steps, a total of 81 meshes and approximately 430 thousand data points remained.

4.2 Implementation

In this case study, a regression-type NN model was adopted as the deterioration prediction model, based on its flexibility in capturing nonlinear and interactive relationships and on the assumption that pavement deterioration progresses as a continuous phenomenon. Although pavement condition in the inspection data is recorded as three discrete classes, these classes have a clear ordinal structure, where larger values indicate more advanced deterioration. Conventional multi-class classification models typically employ a softmax activation function in the output layer, which cannot account for this ordinal relationship. Therefore, to appropriately capture both the ordinal nature of pavement condition and the continuous progression of deterioration, this study constructed a regression-type NN model without an activation function in the output layer. Continuous prediction outputs were then classified into Condition Level 1 if the value was below 1.5, Condition Level 2 if below 2.5, and Condition Level 3 if 2.5 or higher, using a predefined threshold rule.

The NN model consisted of seven layers, with 96 nodes in the input layer and hidden layers comprising 64, 128, 256, 128, 64, and 32 nodes, respectively (see Fig. 1). The ReLU activation function was applied to all hidden layers. The model takes the explanatory variables listed in Table 1 as input and outputs the pavement condition as a continuous value. The dataset was divided into training and test sets at a ratio of 80% to 20%, and the training data were shuffled at the beginning of each epoch. The model was trained using the Adam optimizer with a learning rate of 0.001 and mean squared error (MSE) as the loss function. The number of training epochs was set to 100, and the batch size was set to 16. The model was implemented in Python 3.10 using PyTorch 2.6.0 + cu126 and scikit-learn 1.6.1, and, unless otherwise noted, default parameter initialization settings in PyTorch were used. Hyperparameters such as the number of layers, number

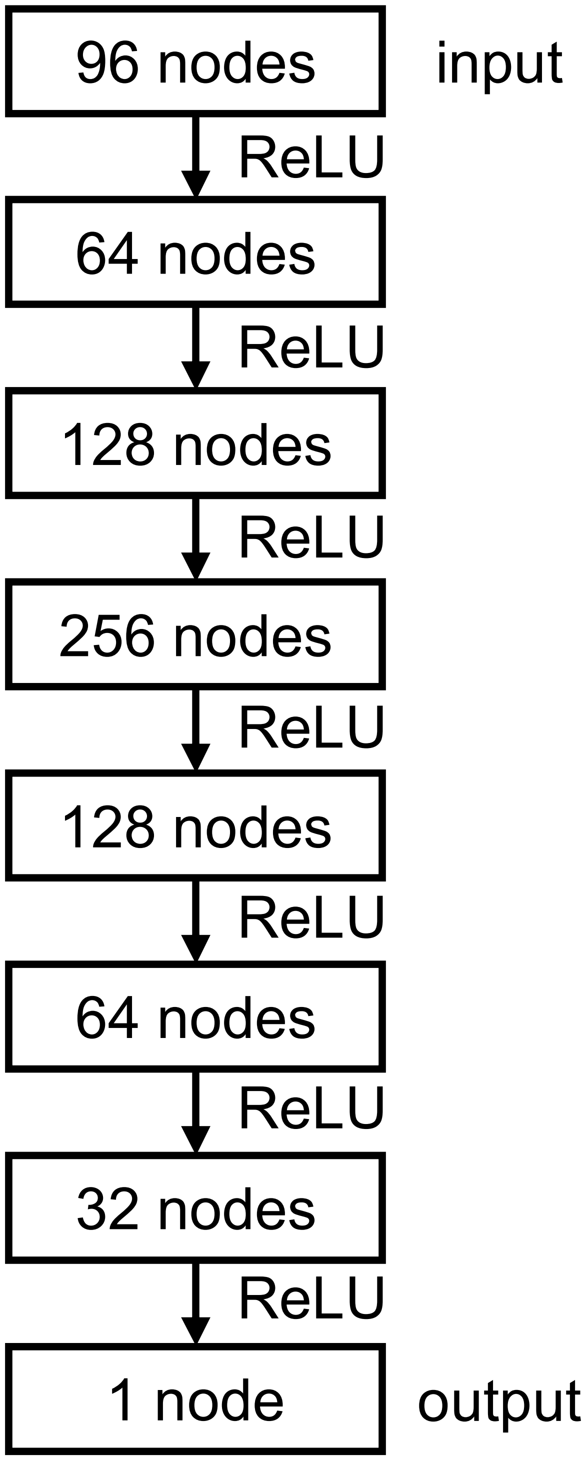

Fig. 1 Neural network architecture

of nodes in each layer, training epochs, and batch size were determined through exploratory preliminary experiments using a small subset of the data, and selected to maximize accuracy based on the above threshold-based classification rule.

As a result of applying the trained model to pavement condition classification, the accuracy was 0.8043, confirming that the proposed framework has sufficient practical precision for assessing pavement condition. Meanwhile, the loss function value (MSE) was 0.1775 (corresponding to an RMSE of 0.4213), which is relatively large. This is because the regression model outputs continuous values that do not necessarily coincide with the integer condition levels, leading to upward or downward deviations, particularly around Condition Levels 1 and 3. However, these deviations are effectively absorbed by the threshold-based rule described above, and thus have little impact on the final classification outcomes. Therefore, the regression-type NN model is deemed effective as a practical deterioration prediction method, as it appropriately captures the continuity of deterioration and the ordinal nature of condition levels while maintaining high classification accuracy.

For the SHAP value computation, 4,000 data points were randomly sampled from the training dataset as background data. Two types of target data sampling were performed: global sampling and region-specific sampling. In global sampling, 4,000 data points were randomly selected from the entire training dataset, while in region-specific sampling, 4,000 data points were drawn separately for each region. DeepSHAP, an algorithm designed for deep learning models, was used to compute SHAP values. DeepSHAP was adapted because it leverages the internal structure of neural networks and efficiently estimates the contribution of each feature to the model output through backpropagation, offering superior computational stability, speed, and scalability. The SHAP analysis was implemented in Python 3.10 using the shap library (version 0.47.1).

4.3 Results and discussion

Figure 2 shows a SHAP summary plot based on global sampling across the entire dataset. In this plot, the color of each dot represents the value of the corresponding explanatory variable, while the x-axis indicates its SHAP value—that is, the degree of influence it has on the predicted deterioration state. The plot displays the top 20 variables with the greatest impact on the model's output, ranked according to the mean absolute SHAP value. Variables shown toward the top have greater relative importance in predicting deterioration.

Figure 2 shows that longer service periods are generally associated with higher positive SHAP values, indicating that deterioration tends to progress with increased time in service—a result that is consistent with intuitive expectations. Usage-related factors such as traffic volume and travel speed have particularly strong effects on predicted deterioration. Structural attributes like pavement

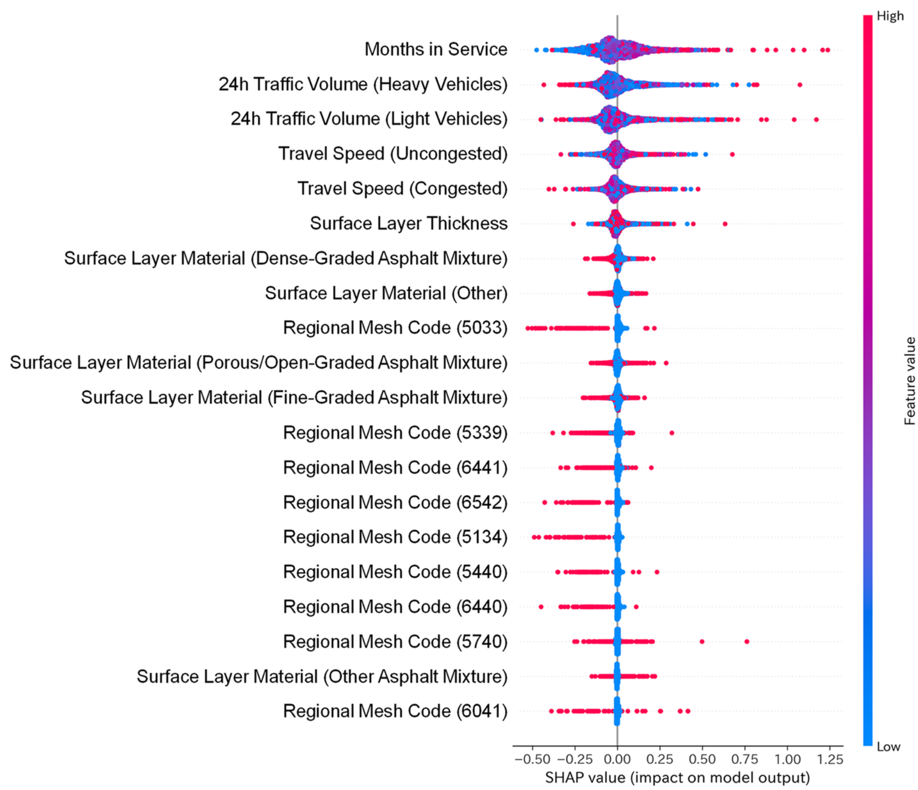

Fig. 2 SHAP summary plot for global sampling

thickness and material follow in importance, with regional location appearing thereafter. These results suggest that, although the impact of regional heterogeneity is smaller than that of primary factors like service duration or traffic volume, it is comparable to the influence of structural conditions such as pavement material.

Next, this study seeks to capture the diversity of deterioration patterns within and between regions based on the distribution of SHAP values. Figures 3, 4, 5 and 6 present the aggregated SHAP values related to regional location, summarized at the primary regional mesh level. These figures show the mean, the first quartile (Q1), the third quartile (Q3), and Pearson's first skewness coefficient for each mesh cell. Meshes that appear blank indicate either the absence of nationally administered highways or that the number of valid data points is fewer than 6,000. For reference, a map of Japan is provided in Fig. 7.

Figure 3 shows that there are noticeable biases in the spatial distribution of mean SHAP values, suggesting that the shape of the SHAP value distributions differs across regions. To more clearly evaluate the distributional characteristics of SHAP values within each region, Q1, Q3, and Pearson's first skewness coefficient were calculated for each region (Figs. 4, 5 and 6). Pearson's first skewness coefficient is defined as follows:

$$Pearson'sFirstSkewnessCoefficient := \frac{mean - mode}{standard\_deviation}$$

Positive values indicate a distribution with a longer right tail, while negative values indicate a longer left tail.

Figure 3 shows that mean SHAP values are higher in certain areas such as around the Tsugaru Strait, Niigata, and Osaka, indicating that these regions are more prone to deterioration. However, the differences between regions are more complex. Figure 4 shows that the first quartile values are low in most regions, suggesting that it tends to be rare for an entire regional grid to exhibit uniformly high deterioration risk. In contrast, Fig. 5 demonstrates that the third quartile values vary greatly between regions, indicating substantial differences in the proportion of highly deteriorated sections within each region. Furthermore, Fig. 6 reveals that in western Japan, many regions exhibit positive skewness, whereas in eastern Japan, negative skewness is more dominant. Integrating these findings, it can be inferred that in western Japan, deterioration tends to be concentrated in a few critical areas, while most other sections remain in relatively good condition (e.g., Fig. 8). Conversely, in eastern Japan, particularly northern Japan, although some sections remain sound, deterioration is more widespread overall (e.g., Fig. 9). These results indicate that there is considerable

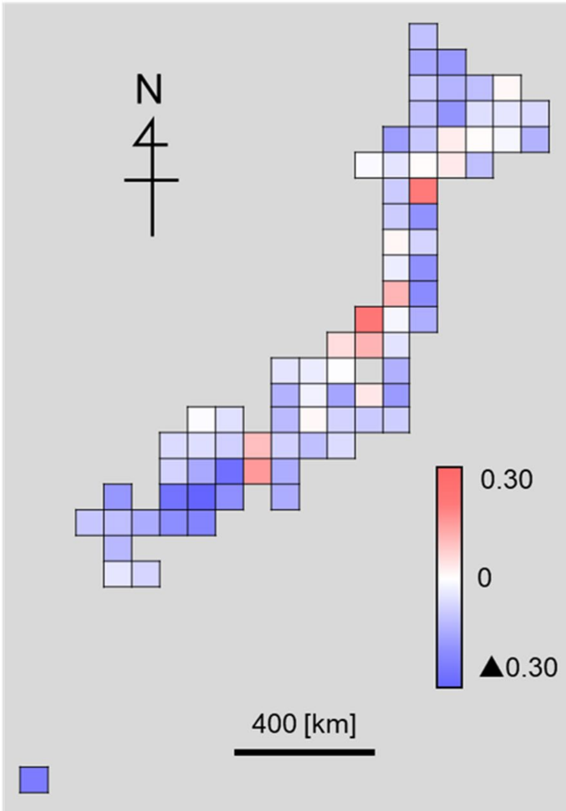

Fig. 3 Spatial distribution of mean SHAP values by mesh

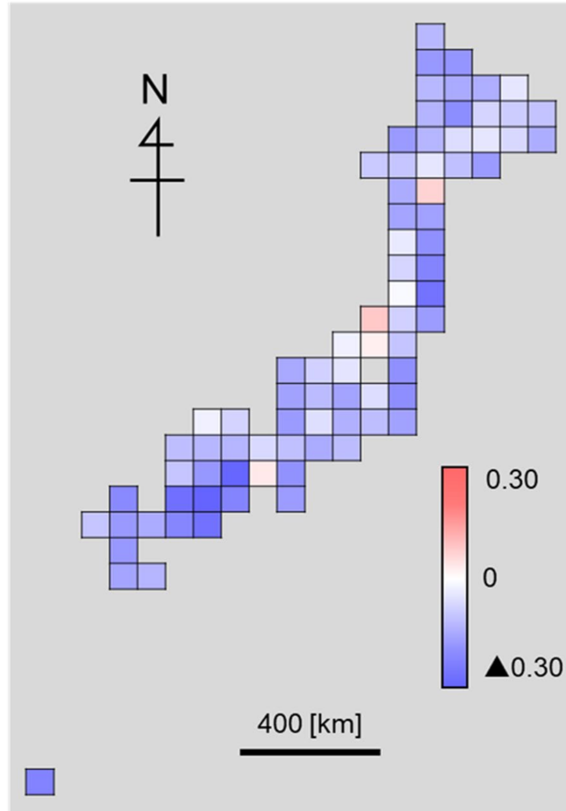

Fig. 4 Spatial distribution of Q1 SHAP values by mesh

diversity in susceptibility to deterioration within each region, and that the spatial distribution of deterioration patterns differs significantly between regions. In particular, third quartile values are notably high in northern Japan and along the Sea of Japan coast, suggesting a potential relationship with snowfall. At the same time, Osaka appears as an outlier: despite limited snowfall, deterioration speed is relatively high. A plausible explanation is usage-related stress. We hypothesize that (1) heavy-vehicle intensity is high due to proximity to major ports and logistics corridors (e.g., Kobe/Osaka), (2) recurrent congestion leads to low operating speeds and frequent stop–go loading, and (3) week subgrade conditions near sea or river contribute to the high deterioration speed. To examine these hypotheses, comparative evaluations of multiple factors are required.

Overall, these findings demonstrate that by comparing multiple statistical indicators derived from SHAP distributions, the proposed framework enables a detailed understanding of both intra- and inter-regional heterogeneity in deterioration behavior.

5 Strengths and limitations of the proposed framework

5.1 Strengths of the proposed framework

First, the proposed framework enables the quantification of regional deterioration propensity by visualizing the distribution of SHAP values for each region. This allows for the identification of intra-regional variability and bias in deterioration tendencies, as well as the characterization of inter-regional heterogeneity in deterioration behavior.

Second, the framework is not constrained by any specific deterioration mechanism and is designed to be applicable to the outputs of any deterioration prediction model. Therefore, it can be extended to other infrastructure assets with different deterioration mechanisms by selecting an appropriate prediction model. Furthermore, in the present case study, the NN model was employed as a general deterioration prediction model, as it does not assume any particular deterioration mechanism.

Third, both the NN model and DeepSHAP used in this case study are based on mini-batch learning and background data sampling, which ensures that computational

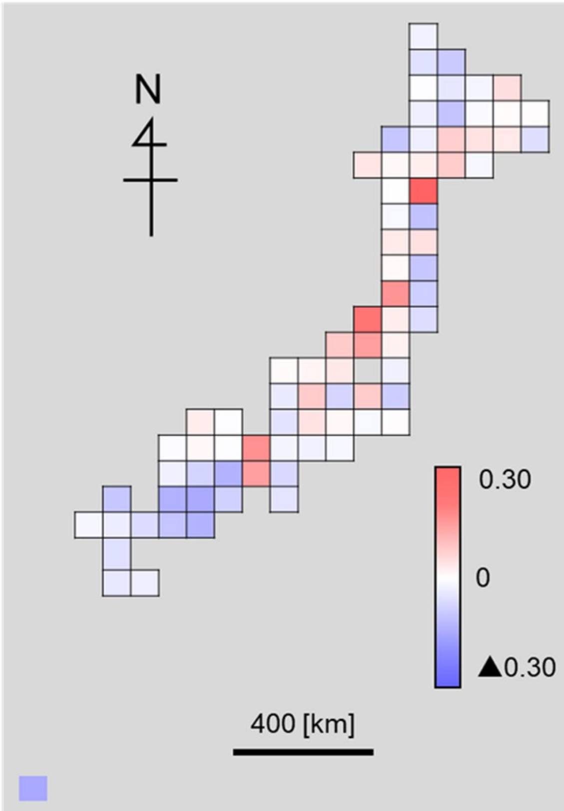

Fig. 5 Spatial distribution of Q3 SHAP values by mesh

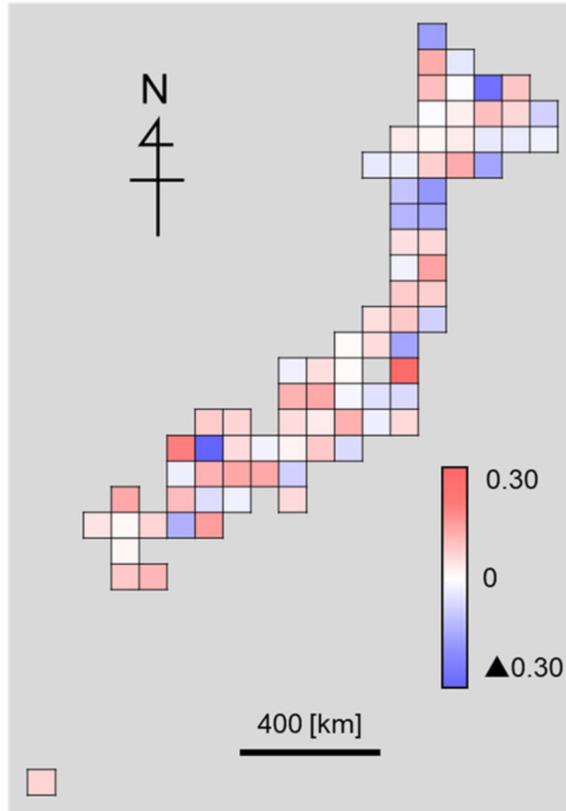

Fig. 6 Spatial distribution of SHAP value skewness by mesh

costs do not increase exponentially as the dataset size expands. On the contrary, an increase in data volume may enhance the generalization capability of the model and improve the accuracy of regional pattern extraction based on SHAP values.

5.2 Limitations of the proposed framework

This study has several limitations, which are discussed below.

First, there are methodological limitations inherent to SHAP itself. The exact computation of SHAP values requires substantial computational cost, so in practice, sampling-based data reduction and approximation methods are generally employed. However, the assumption of independence among explanatory variables, which is often made when adopting approximation methods, is a strong one and may lead to overestimation or underestimation of SHAP values. In the future, it will be important to refine this approach by incorporating statistical techniques commonly used in causal inference, such as propensity scores, to control confounding factors and rigorously analyze the influence of each explanatory variable on deterioration while mitigating bias.

Fig. 7 Map of Japan

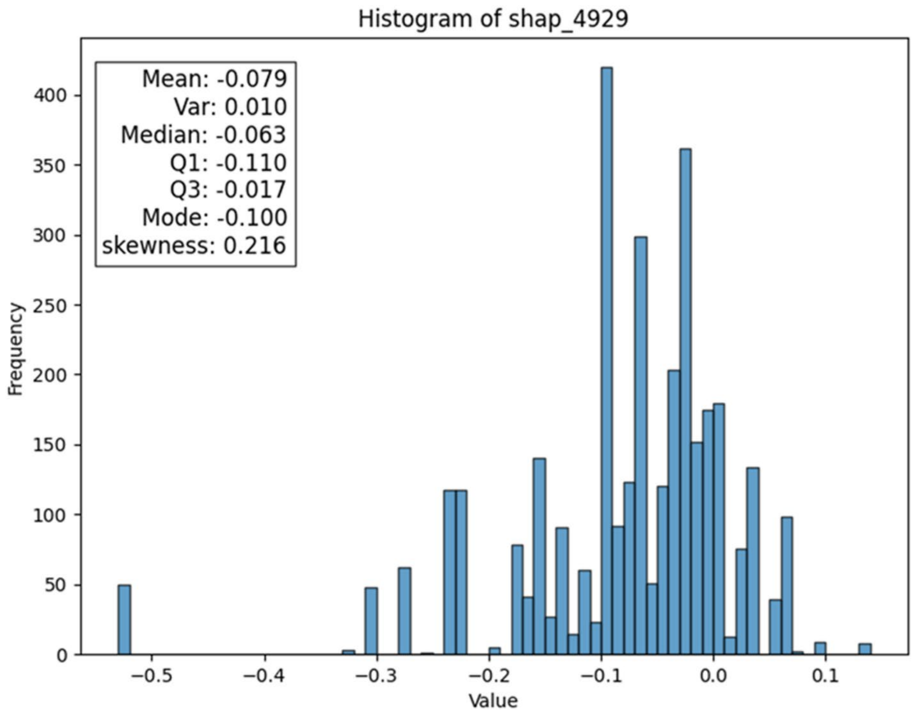

Fig. 8 Histogram of SHAP values for mesh 4929

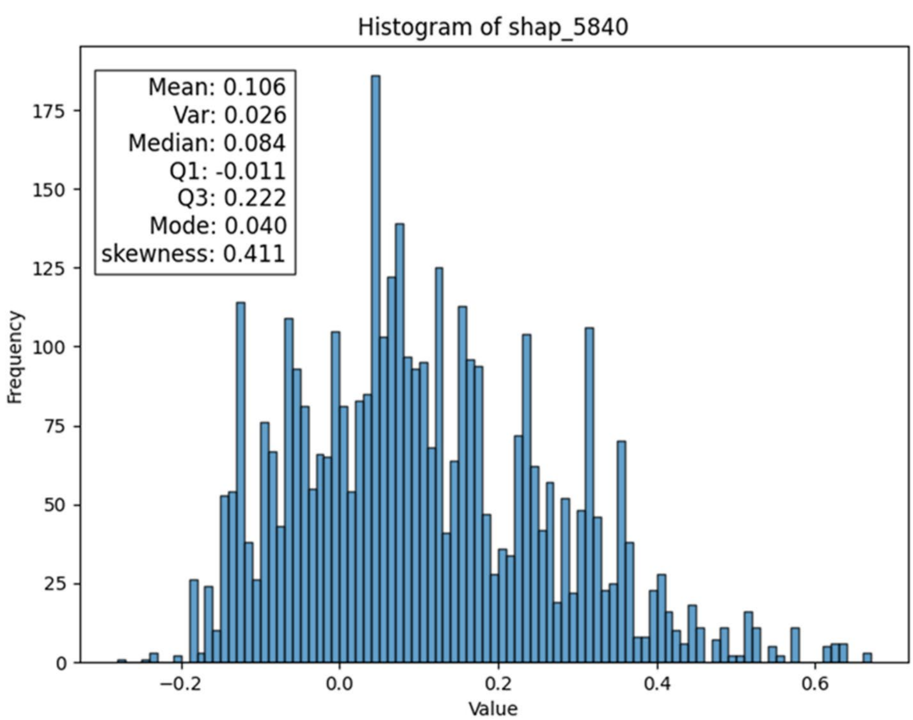

Fig. 9 Histogram of SHAP values for mesh 5840

Second, there are limitations related to data availability. The pavement inspection data used in this study vary in coverage and accuracy across regions and routes, meaning that uniformly high-resolution data were not available nationwide. In data-scarce cases, the region sample size is small and thus the SHAP estimates become high-variance and prone to sampling bias, which reduces generalizability of the proposed framework. Therefore, improving the quantity and quality of infrastructure data, especially in underrepresented areas, remains a key challenge for practical application.

Third, there are limitations in the conceptual treatment of regional heterogeneity. In this study, regional dummy variables were used as proxies for location-based effects. However, further analysis incorporating climate indicators, traffic data, or construction methods may provide deeper insights into the causal structure of regional variation.

Addressing these limitations in future research will be important for enhancing the robustness and applicability of the proposed framework.

6 Conclusion

This study proposed a framework for evaluating regional heterogeneity in infrastructure deterioration by applying SHAP to a deterioration prediction model based on a neural network. By visualizing the contribution of each explanatory variable to predicted deterioration in the form of region-specific distributions, the framework allows for detailed and quantitative identification of differences in deterioration behavior both within and across regions.

A case study using nationwide pavement inspection data on national highways in Japan was conducted to assess the effectiveness of the proposed framework. The results revealed that regional location has a meaningful, non-negligible impact—smaller than primary factors such as time in service or traffic volume, but comparable to structural attributes such as pavement material. Furthermore, analysis of SHAP distributions using statistical indicators such as mean, quartiles, and skewness revealed that there is diversity in deterioration tendencies within each region and that the distribution of these tendencies varies significantly across regions.

The proposed framework enables infrastructure managers to gain insight into how regional characteristics interact with other factors to influence deterioration. This insight is expected to support more regionally tailored and effective maintenance planning and policy development.

In the future, this framework may be extended to other types of infrastructure, such as bridges or tunnels, and combined with more advanced modeling techniques. Further work should also explore the integration of external regional data (e.g., climate conditions, construction standards) to enhance interpretability and support evidence-based asset management strategies.

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Authors' contributions
Atsushi Sugama conceived and supervised the study, led the development of the methodology and software, acquired funding, administered the project, and wrote the original draft. Kenshin Kuwahara contributed to the software development and visualization. Yuto Nakazato contributed to the methodology and formal analysis. All authors were involved in the validation of the results and editing of the manuscript. All authors read and approved the final manuscript.

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Data availability
The datasets generated and analyzed during the current study are available from the corresponding author, Atsushi SUGAMA, upon reasonable request. In addition, this study also utilized publicly available data from the Ministry of Land, Infrastructure, Transport and Tourism [10, 11]. The availability of these external datasets is subject to the terms and restrictions of the original data provider.

Declarations

Competing interests
The author, Atsushi Sugama, is one of the guest editors for Special Issue on the subject of "Resilience and Intelligence on Urban Lifeline: Selected Papers from the 2nd International Symposium on Urban Lifeline" of Urban Lifeline and was not involved in the editorial review, or the decision to publish this article. All authors declare that there are no other competing interests.

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References

1. Morikawa H (2004) Current status and challenges of deterioration prediction technology. Concrete Journal (Japan Concrete Institute) 42(5):52–62. [in Japanese] https://doi.org/10.3151/coj1975.42.5_52
2. Kaito K, Matsumoto K, Kamata T, Kitano Y, Yamanaka A (2023) Simulation of sewer pipe condition using spatial mapping and extraction of rehabilitation and renewal areas. J Japan Soc Civil Eng 79(1):22–00111. [in Japanese] https://doi.org/10.2208/jscejj.22-00111
3. Kaito K, Kobayashi K, Aoki K, Yoneyama H, Matsumoto K (2021) Evaluation of pavement deterioration rate using spatial mapping: practice in Myanmar. J Civil Eng Pract (JSCE) 77(1):84–100. [In Japanese] https://doi.org/10.2208/jscejppce.77.1_84
4. Kleiner Y, Rajani B (2001) Comprehensive review of structural deterioration of water mains: statistical models. Urban Water 3(3):131–150. https://doi.org/10.1016/S1462-0758(01)00033-4
5. Frangopol DM, Kallen MJ, Noortwijk JMV (2004) Probabilistic models for life-cycle performance of deteriorating structures: review and future directions. Prog Struct Eng Mater 6(4):197–212

6. Avci O, Abdeljaber O, Kiranyaz S, Hussein M, Gabbouj M, Inman DJ (2021) A review of vibration-based damage detection in civil structures: from traditional methods to machine learning and deep learning applications. Mech Syst Signal Process 147:107077. https://doi.org/10.1016/j.ymssp.2020.107077
7. Llopis-Castelló D, García-Segura T, Montalbán-Domingo L, Sanz-Benlloch A, Pellicer E (2020) Influence of pavement structure, traffic, and weather on urban flexible pavement deterioration. Sustainability 12(22):9717. https://doi.org/10.3390/su12229717
8. Yu B, Xing H, Yan J, Li Y (2024) Small-scale, large impact: utilizing machine learning to assess susceptibility to urban geological disasters—a case study of urban road collapses in Hangzhou. Bull Eng Geol Environ 83(11):1–19. https://doi.org/10.1007/s10064-024-03931-3
9. Lundberg S, Lee S (2017) A unified approach to interpreting model predictions. arXiv preprint: 1705.07874. https://doi.org/10.48550/arXiv.1705.07874
10. Ministry of Land, Infrastructure, Transport and Tourism (n.d.) National road facility inspection database. https://road-structures-db.mlit.go.jp/
11. Ministry of Land, Infrastructure, Transport and Tourism (2023) 2021 national road and street traffic survey. https://www.mlit.go.jp/road/census/r3/
12. Ministry of Land, Infrastructure, Transport and Tourism (2016) Pavement inspection guidelines. https://www.mlit.go.jp/road/sisaku/yobohozen/tenken/yobo28_10.pdf

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